

Chronic disease and infant nutrition: is it significant to public health?



OBJECTIVE: To assess the public health significance of premature weaning of infants from breast milk on later-life risk of chronic illness. DESIGN: A review and summary of recent meta-analyses of studies linking premature weaning from breast milk with later-life chronic disease risk is presented followed by an estimation of the approximate exposure in a developed Western country, based on historical breast-feeding prevalence data for Australia since 1927. The population-attributable proportion of chronic disease associated with current patterns of artificial feeding in infancy is estimated. RESULTS: After adjustment for major confounding variables, current research suggests that the risks of chronic disease are 30-200 % higher in those who were not breast-fed compared to those who were breast-fed in infancy. Exposure to premature weaning ranges from 20 % to 90 % in post-World War II age cohorts. Overall, the attributable proportion of chronic disease in the population is estimated at 6-24 % for a 30 % exposure to premature weaning. CONCLUSIONS: Breast-feeding is of public health significance in preventing chronic disease. There is a small but consistent effect of premature weaning from breast milk in increasing later-life chronic disease risk. Risk exposure in the Australian population is substantial. Approximately 90 % of current 35-45-year-olds were weaned from breast-feeding by 6 months of age. Encouraging greater duration and exclusivity of breast-feeding is a potential avenue for reducing future chronic disease burden and health system costs.